The modern web

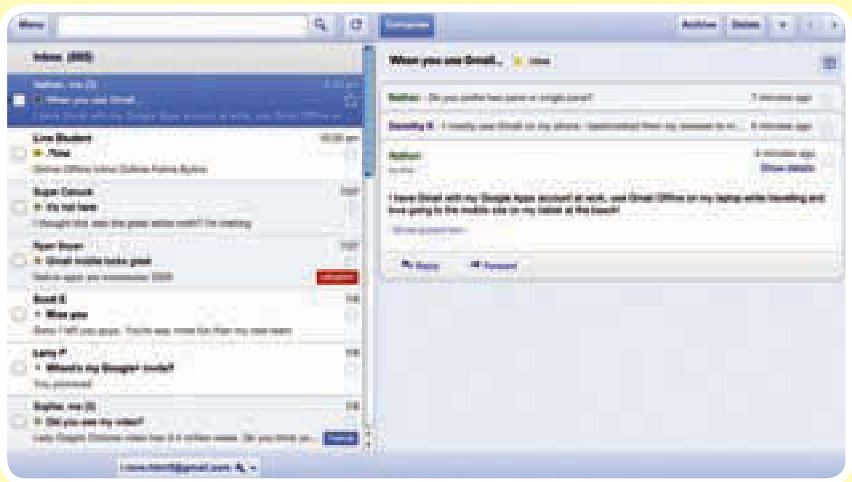
ONE PLATFORM to rule them all

<Rapid improvement in HTML5 and CSS3 has triggered browser vendors to embrace latest standards to stay competitive> **By Kshitij Sobti**

One of the more interesting things about the latest iteration of HTML is that it intends to make the whole process of rendering a web page fool proof. One of the primary limitation of the previous generation of standards has been the way browsers handle pages that aren't standards-compliant. This, unfortunately, varies across different browsers. This necessitated the use of tricks and browser-specific code that made web development an odious task. HTML5 obviates all that by defining the rules for rendering such erroneous pages, as part of the standard. With new generation standards in place, and assuming that the browser is standards compliant, you should see the exact same page replicated across all browsers!

Status quo

If you feel rampant growth in adaptibility of mobile devices is a new trend and hence standards-compliance of mobile browsers would be a major concern, you'd be surprised to know that it's the other way around. Most mobile OSs today ship with a modern browser that supports the latest web standards. In fact, browser statistics reveal that mobile devices are better prepared for newer standards.



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All major smartphone platforms, iOS, Android, BlackBerry, MeeGo, webOS, and bada include browsers with good support for emerging standards.The only platform lagging in support is Windows Phone; hopefully not for long.

It is safe to assume that the device capable of running your application also has a standards-compliant browser. More so on mobile platforms than on desktops actually.

Another important fact to consider is that unlike desktop platforms, mobile OSs offer a limited set of APIs that restrict access to the device. The web offers an even more restricted set. However, if you can create the same experience using web APIs as the device APIs, the web is obviously a better choice!

The web is also incredibly open, and that is something to embrace. The App Store (or any other store) could block your application based on their arbitrary rules, but they cannot block your web applications without crippling the device's browser.

A number of platforms are now embracing web standards as a means to create applications. MeeGo's successor, Tizen, is an HTML5-based platform, webOS entirely allows web applications, iOS allows pinning web apps as normal apps, and Android too can have bookmarks on the home screen.

The web's expanding scope

You may wonder though, can you really make a full-fledged application that runs in the browser?

Any programming language will offer you a syntax for describing logic, along with a standard library that exposes some basic system functionality. C and C++ are preferred over other languages when it comes to system programming as they offer unprecedented control over the native hardware. Although for most tasks it is easier to implement working code in many other languages. While JavaScript makes is incredibly simple to add some logic to web pages, it is limited on several other fronts. This was for a good reason of course; but, it also limited what was possible inside a browser.

This wasn't the only barrier. Imagine something like Wikipedia at a time when the internet was still new, slow, and expensive. How convenient would it have been for people to access and edit an encyclopaedia over a faltering dial-up connection, while paying large sums of money for the privilege? No, at that time even something as simple as a text-based encyclopaedia might have seemed unsuitable for the web.

The internet now reaches more people, it has become ubiquitous, cheap, and fast, and the Wikipedia model is thriving. Some now look at the increasing prevalence of web applications with disdain, because apps don't fit the browser paradigm, but that is changing.

Today it seems odd to use a native email client when it can all be done in a browser. Even though WebMail has all the failings of a website, it is slower than a native client, it is not accessible offline, and it isn't as well integrated with your system. These disadvantages are fast disappearing